

Problem Statement: Text-Based Adventure Game Engine

Interactive storytelling and role-playing games (RPGs) have long been a significant domain in software development, combining narrative design, user interaction, and system architecture. Text-based adventure games, in particular, offer a unique environment where players engage with the game world through typed commands, relying on imagination and logical reasoning rather than graphical interfaces. Despite their conceptual simplicity, developing a robust and extensible text-based adventure engine presents several technical and design challenges, especially when aiming to support both gameplay and content creation within a unified system.

One of the primary challenges lies in designing a flexible and scalable architecture capable of representing complex game worlds. A typical adventure game consists of interconnected rooms, diverse items, non-player characters (NPCs), and dynamic events. These elements must interact seamlessly while maintaining consistency in game state and behavior. Without a well-structured object-oriented design, managing relationships between entities—such as player interactions with items or NPC-driven events—can quickly become unmanageable and error-prone. Additionally, ensuring extensibility for future features, such as new commands or game mechanics, requires careful architectural planning.

Another significant challenge is interpreting and processing user input. Unlike graphical interfaces with predefined actions, text-based games rely on natural language-like commands (e.g., “go north,” “take sword,” “talk to merchant”). Designing a reliable command parser that can handle variations in user input, map commands to in-game actions, and provide meaningful feedback is a non-trivial task. Poor input handling can lead to user frustration and reduced engagement, making it essential to implement a robust command-processing mechanism.

Game state management is also a critical concern. The system must track player progress, inventory, room states, NPC interactions, and event triggers across multiple turns. This includes handling conditional logic, such as unlocking new areas, triggering dialogues, or initiating combat scenarios based on player actions. Ensuring consistency and persistence of game state, particularly through save/load functionality, adds another layer of complexity. Serialization mechanisms must accurately capture and restore the entire game world without data loss or corruption.

In addition to gameplay, there is a growing need for tools that enable users to create and customize their own game worlds. Many existing engines lack intuitive authoring capabilities, requiring developers to manually code game content. This creates a barrier for non-programmers or designers who wish to focus on storytelling and world-building. Providing an integrated game editor that allows users to define rooms, items, NPCs, and dialogues in a structured and user-friendly manner is therefore a key requirement.

This project aims to address these challenges by developing a Text-Based Adventure Game Engine that combines a powerful runtime system with integrated authoring tools. The engine will employ object-oriented design principles, utilizing abstractions, design patterns such as Command, Factory, and Observer, and modular components to ensure flexibility and maintainability. It will support a rich world model, command parsing, turn-based execution, and dynamic event handling.

Furthermore, the system will include features for saving and loading game states via JSON serialization, as well as debugging and testing tools to assist developers and content creators. Within the constraints of a 10-week development period, the project will deliver a functional prototype demonstrating core gameplay mechanics, extensible architecture, and basic world-building capabilities. By bridging the gap between game engine design and content creation, this project aims to provide both an engaging user experience and a practical platform for interactive storytelling development.